PingSheng Kao

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SUMMARY

ML Data Scientist with 5+ years of experience in multimodal and agent evaluation, auto-judge design, human-AI calibration, embedding-based training-data curation, and targeted vision-model fine-tuning. Leads cross-functional evaluation initiatives from research questions through internal production measurement.

EXPERIENCE

ML Data Scientist - Apple July 2020 - Present
Santa Clara, California

- Developed an auditable, reference-based multi-stage LLM evaluator and expert-calibrated data workflow for a multimodal health AI product; deployed internally for performance reporting and failure discovery and reached 90% expert alignment on a 40-example gold set.
- Led cross-functional translation from user-research needs to engineering requirements and Data Operations labeling instructions, coordinating LLM-as-judge evaluation with human-in-the-loop validation.
- Built a simulated-user agent, agent-as-judge workflow, targeted scenario generators, and user-centered metrics for an open-ended multimodal conversational agent; findings changed subsequent development direction.
- Contributed to large-scale embedding-index experiments that curated an internal segmentation training dataset aligned with Meta's Segment Anything research and delivered the dataset to a foundation-model team.
- Owned designated evaluators and shipped versioned updates while scaling LLM- and agent-as-judge runs on an existing internal batch platform, debugging throughput and rate-limit issues and publishing dashboards.
- Developed scene-graph evaluation for OCR, counting, attribute, and spatial-reasoning failures, and fine-tuned vision models on targeted failure slices.

RESEARCH CONTRIBUTION

GH-ESD: Grounded Hypothesis-Driven Error Slice Discovery for Instance-Level Vision Tasks 2025
Experiment contributor (credited as Sam Kao) - accepted to ECCV - arXiv:2512.24592

- Contributed experiments to ECCV-accepted research on grounded, hypothesis-driven error-slice discovery for instance-level vision tasks.

ADDITIONAL EXPERIENCE

Algorithm and Architecture Intern - Deep Learning Compression - Ambarella June 2019 - August 2019
Santa Clara, California

- Developed a saliency-based neural-network quantization heuristic that reduced an exponential search formulation to a linear-time method, and proposed a mixed-precision algorithm integrated into company software.

EDUCATION

M.S. in Computer Science and Engineering - University of Michigan, Ann Arbor 2018 - 2020
B.S. in Electronic Engineering - National Taiwan University 2013 - 2017
GPA: 3.9/4.0 (B.S.)

SKILLS

Programming: Python, SQL, C++, JavaScript
Multimodal & Agents: LLM-as-Judge, Agent-as-Judge, User Simulation, Scene Graphs, Tool-Call Evaluation
ML & Data: PyTorch, Vision-Model Fine-Tuning, Training-Data Curation, Embedding Indexing, Clustering
Evaluation: Benchmark Design, Human-AI Calibration, Synthetic Scenarios, Failure Analysis, DSPy